Experiment 3: Probability Distributions and Bayes' Theorem

Aim

To generate and visualise standard probability distributions, calculate the conditional probability of disease using Bayes' theorem, verify the result through simulation, and study the effect of disease prevalence on posterior probability.

Mathematical Formulae

1. Bayes' theorem

$$P(D \mid +) = \frac{P(+ \mid D)P(D)}{P(+)}$$

2. Total probability of a positive result

$$P(+) = P(+ \mid D)P(D) + P(+ \mid \bar{D})P(\bar{D})$$

3. Bernoulli distribution

$$P(X = x) = p^x(1-p)^{1-x}$$

4. Binomial distribution

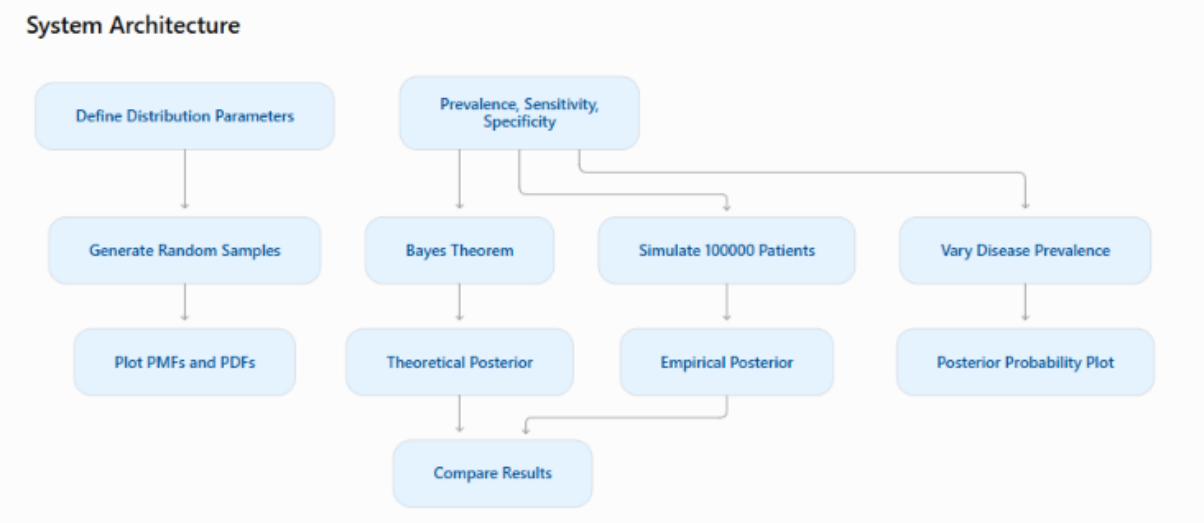
$$P(X = k) = \binom{n}{k} p^k (1-p)^{n-k}$$

5. Poisson distribution

$$P(X = k) = \frac{e^{-\lambda}\lambda^k}{k!}$$

6. Gaussian probability density

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$$



Algorithm / Procedure

1. Import NumPy, SciPy and Matplotlib.
2. Generate samples from Bernoulli, Binomial, Poisson, Uniform and Gaussian distributions.
3. Plot the empirical distributions and theoretical PMFs/PDFs.
4. Define prevalence as 1%, sensitivity as 95% and specificity as 90%.
5. Calculate the probability of disease given a positive test using Bayes' theorem.
6. Simulate 100,000 patients and estimate the empirical posterior probability.
7. Vary prevalence from 0.1% to 20% and plot the posterior probability.

Python Program

```
import numpy as np
import matplotlib.pyplot as plt
from scipy.stats import bernoulli, binom, poisson, uniform, norm

np.random.seed(42)

# 1. Generate samples from five distributions
samples = {
  "Bernoulli": bernoulli.rvs(p=0.4, size=1000),
  "Binomial": binom.rvs(n=10, p=0.4, size=1000),
  "Poisson": poisson.rvs(mu=4, size=1000),
```

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    "Uniform": uniform.rvs(loc=0, scale=1, size=1000),
    "Gaussian": norm.rvs(loc=0, scale=1, size=1000)
}

# Plot the sample distributions
fig, axes = plt.subplots(2, 3, figsize=(13, 7))
axes = axes.flatten()

for i, (name, data) in enumerate(samples.items()):
    if name in ["Bernoulli", "Binomial", "Poisson"]:
        values, counts = np.unique(data, return_counts=True)
        axes[i].bar(values, counts / len(data), alpha=0.7)

        # Theoretical probability mass function
        if name == "Bernoulli":
            theoretical = bernoulli.pmf(values, p=0.4)
        elif name == "Binomial":
            theoretical = binom.pmf(values, n=10, p=0.4)
        else:
            theoretical = poisson.pmf(values, mu=4)

        axes[i].plot(values, theoretical, "ro", label="Theoretical PMF")
        axes[i].set_ylabel("Probability")

    else:
        axes[i].hist(data, bins=30, density=True, alpha=0.7)
        x = np.linspace(data.min(), data.max(), 200)

        if name == "Uniform":
            axes[i].plot(x, uniform.pdf(x, loc=0, scale=1),
                         label="Theoretical PDF")
        else:
            axes[i].plot(x, norm.pdf(x, loc=0, scale=1),
                         label="Theoretical PDF")
```

```
        axes[i].set_ylabel("Density")

    axes[i].set_title(name)
    axes[i].legend()

axes[5].axis("off")
plt.tight_layout()
plt.show()

# 2. Bayes' theorem
prevalence = 0.01
sensitivity = 0.95
specificity = 0.90

p_disease = prevalence
p_positive_given_disease = sensitivity
p_positive_given_no_disease = 1 - specificity

p_positive = (
    p_positive_given_disease * p_disease
    + p_positive_given_no_disease * (1 - p_disease)
)

posterior = (
    p_positive_given_disease * p_disease / p_positive
)

print("\nBayes' Theorem")
print("P(Disease):", prevalence)
print("P(Positive):", round(p_positive, 5))
print("P(Disease | Positive):", round(posterior, 5))
print("Posterior Percentage:", round(posterior * 100, 2), "%")
```

```
# 3. Simulate 100,000 patients
N = 100_000

disease = np.random.rand(N) < prevalence

# Positive test for patients with disease
positive_if_disease = np.random.rand(N) < sensitivity

# Positive test for patients without disease
positive_if_healthy = np.random.rand(N) < (1 - specificity)

positive = np.where(
  disease, positive_if_disease, positive_if_healthy
)

number_positive = positive.sum()

if number_positive > 0:
  empirical = disease[positive].mean()
  print("\nSimulation Results")
  print("Total patients:", N)
  print("Positive tests:", number_positive)
  print("Empirical P(Disease | Positive):",
      round(empirical, 5))
  print("Theoretical P(Disease | Positive):",
      round(posterior, 5))

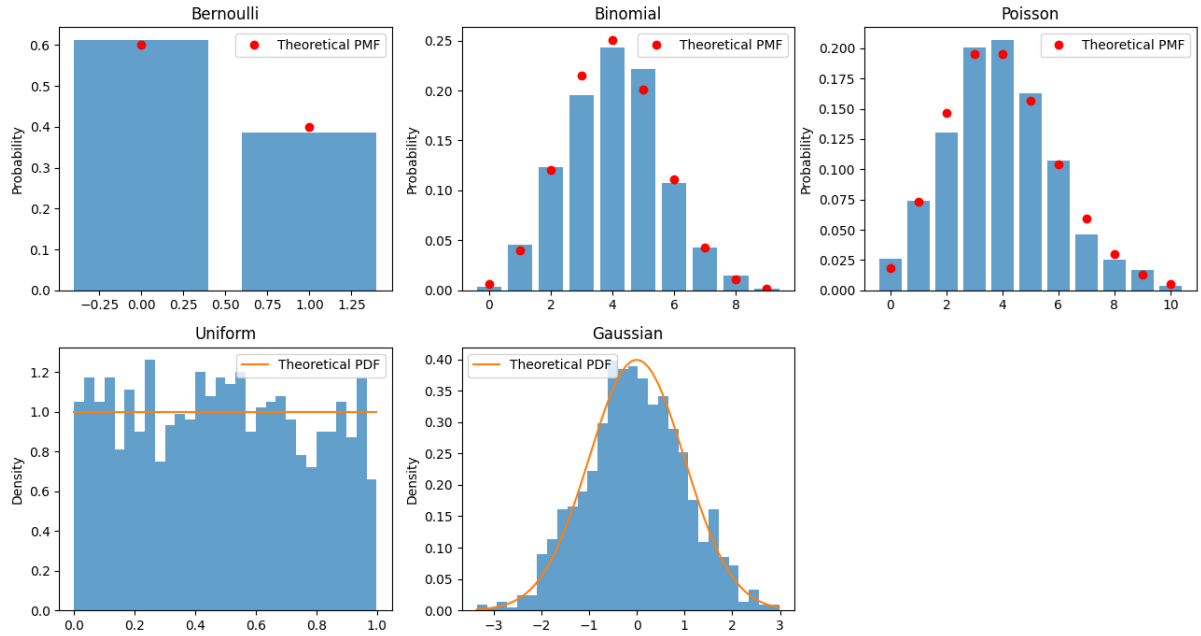
# 4. Posterior probability for varying prevalence
prevalences = np.linspace(0.001, 0.20, 200)

posteriors = (
  sensitivity * prevalences
  / (
    sensitivity * prevalences
```

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        + (1 - specificity) * (1 - prevalences)
    )
)

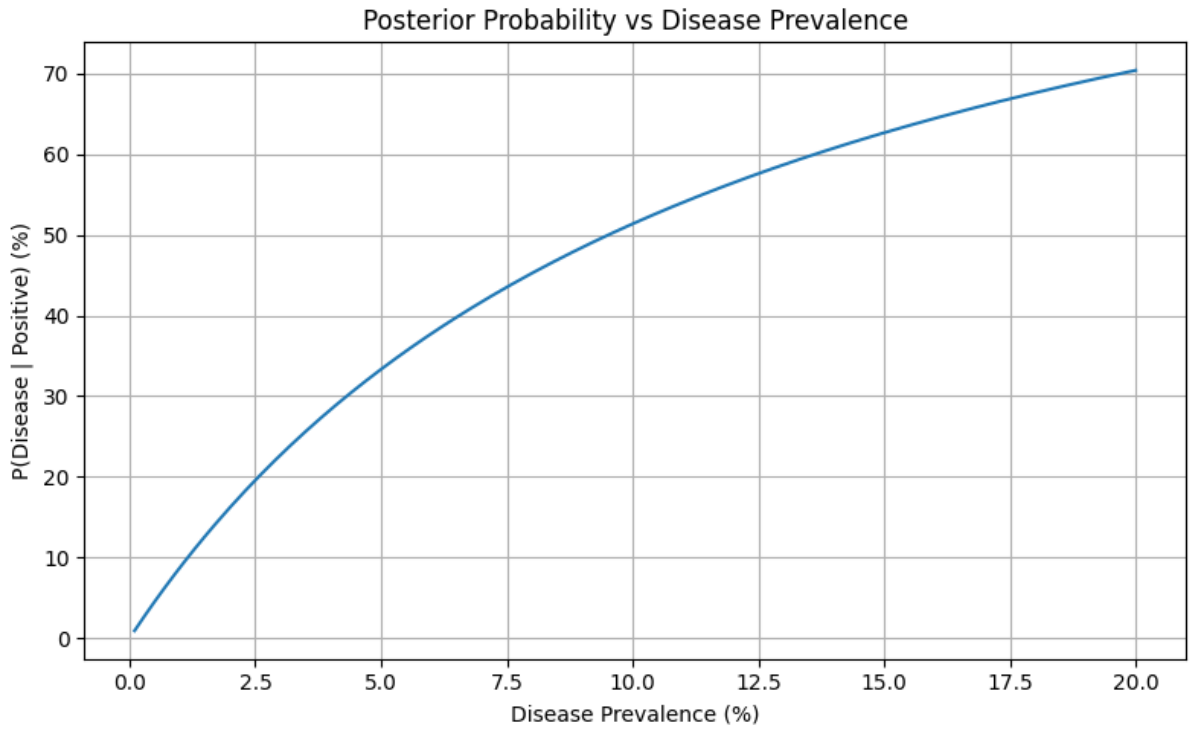
plt.figure(figsize=(8, 5))
plt.plot(prevalences * 100, posteriors * 100)
plt.xlabel("Disease Prevalence (%)")
plt.ylabel("P(Disease | Positive) (%)")
plt.title("Posterior Probability vs Disease Prevalence")
plt.grid(True)
plt.tight_layout()
plt.show()
```

Output




```
Bayes' Theorem
P(Disease): 0.01
P(Positive): 0.1085
P(Disease | Positive): 0.08756
Posterior Percentage: 8.76 %

Simulation Results
Total patients: 100000
Positive tests: 10867
Empirical P(Disease | Positive): 0.08843
Theoretical P(Disease | Positive): 0.08756
```



For the given values:

$$P(D) = 0.01, \quad P(+ \mid D) = 0.95$$

$$P(+ \mid \bar{D}) = 1 - 0.90 = 0.10$$

$$P(D \mid +) = \frac{0.95(0.01)}{0.95(0.01) + 0.10(0.99)}$$

$$\boxed{P(D \mid +) \approx 8.76\%}$$

Result

Thus, five standard probability distributions were generated and visualised. Bayes' theorem was used to calculate the probability of disease given a positive test, and the theoretical result was compared with a simulation of 100,000 patients.